

1. WWA: Healthy Habits

CI: The decisions we make affect our habits

LOI:

How animals process and respond to information (causation)

How we establish our eating habits (connection)

Decision making (responsibility)

LP: Open-minded, principled, reflective

ATL: Social, Self-management

- Analyze characteristics of a diet and compares it to own eating habits

Science:

4-LS1-2. Use a model to describe that animals receive different types of information through their senses, process the information in their brain, and respond to the information in different ways.

Literacy:

Writing:

CCSS.ELA-LITERACY.W.4.2

Write informative/explanatory texts to examine a topic and convey ideas and information clearly.

CCSS.ELA-LITERACY.W.4.2.A

Introduce a topic clearly and group related information in paragraphs and sections; include formatting (e.g., headings), illustrations, and multimedia when useful to aiding comprehension.

CCSS.ELA-LITERACY.W.4.2.B

Develop the topic with facts, definitions, concrete details, quotations, or other information and examples related to the topic.

CCSS.ELA-LITERACY.W.4.2.C

Link ideas within categories of information using words and phrases (e.g., another, for example, also, because).

CCSS.ELA-LITERACY.W.4.2.D

Use precise language and domain-specific vocabulary to inform about or explain the topic.

CCSS.ELA-LITERACY.W.4.2.E

Provide a concluding statement or section related to the information or explanation presented.

CCSS.ELA-LITERACY.W.4.7

Conduct short research projects that build knowledge through investigation of different aspects of a topic.

CCSS.ELA-LITERACY.W.4.8

Recall relevant information from experiences or gather relevant information from print and digital sources; take notes and categorize information, and provide a list of sources.

CCSS.ELA-LITERACY.W.4.9

Draw evidence from literary or informational texts to support analysis, reflection, and research.

[Reading:](#)

[RI. 4.1](#)

[RI. 4.4](#)

[RI.4.7](#)

Math:

CCSS.MATH.CONTENT.4.NBT.A.1

Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right. For example, recognize that $700 \div 70 = 10$ by applying concepts of place value and division.

CCSS.MATH.CONTENT.4.NBT.A.2

Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. Compare two multi-digit numbers based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.

CCSS.MATH.CONTENT.4.NBT.A.3

Use place value understanding to round multi-digit whole numbers to any place.

CCSS.MATH.CONTENT.4.NBT.B.4

Fluently add and subtract multi-digit whole numbers using the standard algorithm.

(Work on symmetry using various objects that humans use and consume including human body)

CCSS.MATH.CONTENT.4.G.A.3

Recognize a line of symmetry for a two-dimensional figure as a line across the figure such that the figure can be folded along the line into matching parts. Identify line-symmetric figures and draw lines of symmetry.

5. HTWW: Biodiversity and Tourism (interactions in biological world) (Westhill Day)

CI: People visit a place for a variety of reasons

LOI:

How tourism is a means of economic development (function)

Connection between biodiversity and tourism (connection)

Cultural diversity (Perspective)

Skills: Research, communication

LP: Knowledgeable, communicators, balanced

- Characteristics of live beings and classifying
- Understand time can be measured by repetitive events and its relationship to live beings
- Analyzes the distribution of water and its importance
- Relates the distribution of climates with the diversity of fauna and flora
- Cultural diversity in Mexico
- Economy in Mexico
- Transportation, communication, tourism in Mexico

Literacy:

Writing: Create Touristic brochure)

CCSS.ELA-LITERACY.W.4.1

Write opinion pieces on topics or texts, supporting a point of view with reasons and information.

CCSS.ELA-LITERACY.W.4.1.A

Introduce a topic or text clearly, state an opinion, and create an organizational structure in which related ideas are grouped to support the writer's purpose.

CCSS.ELA-LITERACY.W.4.1.B

Provide reasons that are supported by facts and details.

CCSS.ELA-LITERACY.W.4.1.C

Link opinion and reasons using words and phrases (e.g., for instance, in order to, in addition).

CCSS.ELA-LITERACY.W.4.1.D

Provide a concluding statement or section related to the opinion presented.

CCSS.ELA-LITERACY.W.4.6

With some guidance and support from adults, use technology, including the Internet, to produce and publish writing as well as to interact and collaborate with others; demonstrate sufficient command of keyboarding skills to type a minimum of one page in a single sitting.

Speaking and Listening

CCSS.ELA-LITERACY.SL.4.4

Report on a topic or text, tell a story, or recount an experience in an organized manner, using appropriate facts and relevant, descriptive details to support main ideas or themes; speak clearly at an understandable pace.

CCSS.ELA-LITERACY.SL.4.5

Add audio recordings and visual displays to presentations when appropriate to enhance the development of main ideas or themes.

CCSS.ELA-LITERACY.SL.4.6

Differentiate between contexts that call for formal English (e.g., presenting ideas) and situations where informal discourse is appropriate (e.g., small-group discussion); use formal English when appropriate to task and situation. (See grade 4 Language standards 1 here for specific expectations.)

[Reading:](#)

[RL.4.1](#)

[RL.4.4](#)

[RL.4.7](#)

4. WWAIPAT: Maps and legends (personal histories, exploration)

CI: People respond to the natural geographical features of the earth

LOI:

How geography determines human settlement (causation)

How geography shapes natural boundaries (form)

The impact of natural disasters (change)

ATL: Thinking, communication

LP: Communicators, thinkers, inquirers

- Mexican territory and political divisions
- Maps to obtain and learn information about territory
- Types of landscapes
- Characteristics of relief forms in Mexico & the spatial distributions dif types of maps
- Characteristics and distribution of population, rural/urban
- Natural disasters

Science:

4-ESS2-2. Analyze and interpret data from maps to describe patterns of Earth's features.

4-ESS1-1. Identify evidence from patterns in rock formations and fossils in rock layers to support an explanation for changes in a landscape over time.

Social Studies:

North America: geography and map skills [4.T1]

Ancient civilizations of North America [4.T2]

The expansion of the United States over time and its regions today 28 [4.T4]

Literacy:

Writing:

CCSS.ELA-LITERACY.W.4.3

Write narratives to develop real or imagined experiences or events using effective technique, descriptive details, and clear event sequences.

CCSS.ELA-LITERACY.W.4.3.A

Orient the reader by establishing a situation and introducing a narrator and/or characters; organize an event sequence that unfolds naturally.

CCSS.ELA-LITERACY.W.4.3.B

Use dialogue and description to develop experiences and events or show the responses of characters to situations.

CCSS.ELA-LITERACY.W.4.3.C

Use a variety of transitional words and phrases to manage the sequence of events.

CCSS.ELA-LITERACY.W.4.3.D

Use concrete words and phrases and sensory details to convey experiences and events precisely.

CCSS.ELA-LITERACY.W.4.3.E

Provide a conclusion that follows from the narrated experiences or events.

CCSS.ELA-LITERACY.W.4.9.A

Apply grade 4 Reading standards to literature (e.g., "Describe in depth a character, setting, or event in a story or drama, drawing on specific details in the text [e.g., a character's thoughts, words, or actions].").

[Reading](#)

[RI.4.2](#)

[RI.4.5](#)

[RI.4.8](#)

Math:

CCSS.MATH.CONTENT.4.MD.A.1

Know relative sizes of measurement units within one system of units including km, m, cm; kg, g; lb, oz.; l, ml; hr, min, sec. Within a single system of measurement, express measurements in a larger unit in terms of a smaller unit. Record measurement equivalents in a two-column table. For example, know that 1 ft is 12 times as long as 1 in. Express the length of a 4 ft snake as 48 in. Generate a conversion table for feet and inches listing the number pairs (1, 12), (2, 24), (3, 36), ...

CCSS.MATH.CONTENT.4.MD.A.2

Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses of objects, and money, including problems involving simple fractions or decimals, and problems that require expressing measurements given in a larger unit in terms of a smaller unit. Represent measurement quantities using diagrams such as number line diagrams that feature a measurement scale.

CCSS.MATH.CONTENT.4.MD.B.4

Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Solve problems involving addition and subtraction of fractions by using information presented in line plots. For example, from a line plot find and interpret the difference in length between the longest and shortest specimens in an insect collection.

6. STP: Environmental issues (relationships between communities, access to equal opportunities, peace and conflict resolution)

CI: Balance mitigates pollution and sustains Earth's resources.

LOI:

Natural resources and how they are obtained (form)

Human responsibility towards sustainability (responsibility)

Causes of pollution (causation)

LP: Caring, balanced, reflective

ATL: Self-management, research

- Assumes responsibility in the prevention and mitigation of environmental issues local and national
- Preservation of biodiversity of national territory
- Distribution and importance of natural resources
- Recognizes causes and effects of water, soil, and air pollution
- How everyday materials are created, their origins and how they are obtained
- Changes of state based on temperature variation

Science:

4-LS1-1. Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction.

4-ESS3-2. Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.

4-ESS2-1. Make observations and/or measurements to provide evidence of the effects of weathering or the rate of erosion by water, ice, wind, or vegetation.

4-ESS3-1. Obtain and combine information to describe that energy and fuels are derived from natural resources and that their uses affect the environment.

Literacy:

Writing:

CCSS.ELA-LITERACY.W.4.1

Write opinion pieces on topics or texts, supporting a point of view with reasons and information.

CCSS.ELA-LITERACY.W.4.1.A

Introduce a topic or text clearly, state an opinion, and create an organizational structure in which related ideas are grouped to support the writer's purpose.

CCSS.ELA-LITERACY.W.4.1.B

Provide reasons that are supported by facts and details.

CCSS.ELA-LITERACY.W.4.1.C

Link opinion and reasons using words and phrases (e.g., for instance, in order to, in addition).

CCSS.ELA-LITERACY.W.4.1.D

Provide a concluding statement or section related to the opinion presented.

Speaking and Listening: (Socratic Seminar, in previous units students have explored each skill need to participate successfully in a SS)

CCSS.ELA-LITERACY.SL.4.1

Engage effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on grade 4 topics and texts, building on others' ideas and expressing their own clearly.

CCSS.ELA-LITERACY.SL.4.1.A

Come to discussions prepared, having read or studied required material; explicitly draw on that preparation and other information known about the topic to explore ideas under discussion.

CCSS.ELA-LITERACY.SL.4.1.B

Follow agreed-upon rules for discussions and carry out assigned roles.

CCSS.ELA-LITERACY.SL.4.1.C

Pose and respond to specific questions to clarify or follow up on information, and make comments that contribute to the discussion and link to the remarks of others.

CCSS.ELA-LITERACY.SL.4.1.D

Review the key ideas expressed and explain their own ideas and understanding in light of the discussion.

CCSS.ELA-LITERACY.SL.4.2

Paraphrase portions of a text read aloud or information presented in diverse media and formats, including visually, quantitatively, and orally.

CCSS.ELA-LITERACY.SL.4.3

Identify the reasons and evidence a speaker provides to support particular points.

[Reading](#)

[RL.4.2](#)

[RL.4.5](#)

Math:

CCSS.MATH.CONTENT.4.NF.C.6

Use decimal notation for fractions with denominators 10 or 100. For example, rewrite 0.62 as 62/100; describe a length as 0.62 meters; locate 0.62 on a number line diagram.

CCSS.MATH.CONTENT.4.NF.C.7

Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions, e.g., by using a visual model.

2. HWEO: (dance as movement and expression)

CI: Cultural Preservation unifies the communication process of ideas, values, and feelings.

LOI:

Movement as energy and expression (Function)

Sound waves and their patterns (Change)

How people preserve culture through movement and sound (perspective)

LP: Risk-taker, balanced, knowledgeable

ATL: Social, self-management

Science:

4-PS3-1. Use evidence to construct an explanation relating the speed of an object to the energy of that object.

4-PS3-3. Ask questions and predict outcomes about the changes in energy that occur when objects collide.

4-PS3-4. Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.* (OK GO video with rube goldburg machine)

4-PS4-1. Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.

4-PS4-3. Generate and compare multiple solutions that use patterns to transfer information.*

Literacy:

Writing:

CCSS.ELA-LITERACY.W.4.3

Write narratives to develop real or imagined experiences or events using effective technique, descriptive details, and clear event sequences.

CCSS.ELA-LITERACY.W.4.3.A

Orient the reader by establishing a situation and introducing a narrator and/or characters; organize an event sequence that unfolds naturally.

CCSS.ELA-LITERACY.W.4.3.B

Use dialogue and description to develop experiences and events or show the responses of characters to situations.

CCSS.ELA-LITERACY.W.4.3.C

Use a variety of transitional words and phrases to manage the sequence of events.

CCSS.ELA-LITERACY.W.4.3.D

Use concrete words and phrases and sensory details to convey experiences and events precisely.

CCSS.ELA-LITERACY.W.4.3.E

Provide a conclusion that follows from the narrated experiences or events.

CCSS.ELA-LITERACY.W.4.9.A

Apply grade 4 Reading standards to literature (e.g., "Describe in depth a character, setting, or event in a story or drama, drawing on specific details in the text [e.g., a character's thoughts, words, or actions].").

Speaking and Listening:

CCSS.ELA-LITERACY.SL.4.4

Report on a topic or text, tell a story, or recount an experience in an organized manner, using appropriate facts and relevant, descriptive details to support main ideas or themes; speak clearly at an understandable pace.

CCSS.ELA-LITERACY.SL.4.5

Add audio recordings and visual displays to presentations when appropriate to enhance the development of main ideas or themes.

CCSS.ELA-LITERACY.SL.4.6

Differentiate between contexts that call for formal English (e.g., presenting ideas) and situations where informal discourse is appropriate (e.g., small-group discussion); use formal English when appropriate to task and situation. (See grade 4 Language standards 1 here for specific expectations.)

[Reading](#)

[RL.4.3](#)

[RL.4.6](#)

[RL.4.9](#)

[RL.4.10](#)

Math:

CCSS.MATH.CONTENT.4.G.A.1

Draw points, lines, line segments, rays, angles (right, acute, obtuse), and perpendicular and parallel lines. Identify these in two-dimensional figures.

3. HWOO: Navigation

- Solar system: eclipse and moon phases, relationship between moon, earth, and sun
- Sources of heat and light and how they are used by human beings

CI: The central role of the solar system on human life

LOI:

Relationship between moon, earth, and sun (connection)

Sources of heat and light and how they are used by human beings (function)

Navigational instruments (function/form)

LP: Thinkers, open-minded, inquirers

ATL: Research, thinking

Science:

4-PS4-2. Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen.

4-PS3-2. Make observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents.

Social Studies:

Early European exploration and conquest [4.T3] (navigation)

Literacy:

Writing:

CCSS.ELA-LITERACY.W.4.2

Write informative/explanatory texts to examine a topic and convey ideas and information clearly.

CCSS.ELA-LITERACY.W.4.2.A

Introduce a topic clearly and group related information in paragraphs and sections; include formatting (e.g., headings), illustrations, and multimedia when useful to aiding comprehension.

CCSS.ELA-LITERACY.W.4.2.B

Develop the topic with facts, definitions, concrete details, quotations, or other information and examples related to the topic.

CCSS.ELA-LITERACY.W.4.2.C

Link ideas within categories of information using words and phrases (e.g., another, for example, also, because).

CCSS.ELA-LITERACY.W.4.2.D

Use precise language and domain-specific vocabulary to inform about or explain the topic.

CCSS.ELA-LITERACY.W.4.2.E

Provide a concluding statement or section related to the information or explanation presented.

CCSS.ELA-LITERACY.W.4.7

Conduct short research projects that build knowledge through investigation of different aspects of a topic.

CCSS.ELA-LITERACY.W.4.8

Recall relevant information from experiences or gather relevant information from print and digital sources; take notes and categorize information, and provide a list of sources.

CCSS.ELA-LITERACY.W.4.9

Draw evidence from literary or informational texts to support analysis, reflection, and research.

CCSS.ELA-LITERACY.W.4.9.B

Apply grade 4 Reading standards to informational texts (e.g., "Explain how an author uses reasons and evidence to support particular points in a text").

[Reading](#)

[RI.4.3](#)

[RI.4.6](#)

[RI.4.9](#)

[RI.4.10](#)

Math:

CCSS.MATH.CONTENT.4.OA.C.5

Generate a number or shape pattern that follows a given rule. Identify apparent features of the pattern that were not explicit in the rule itself. For example, given the rule "Add 3" and the starting number 1, generate terms in the resulting sequence and observe that the terms appear to alternate between odd and even numbers. Explain informally why the numbers will continue to alternate in this way.

(2D representation of solar system)

CCSS.MATH.CONTENT.4.MD.C.5

Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint, and understand concepts of angle measurement:

CCSS.MATH.CONTENT.4.MD.C.5.A

An angle is measured with reference to a circle with its center at the common endpoint of the rays, by considering the fraction of the circular arc between the points where the two rays intersect the circle. An angle that turns through $\frac{1}{360}$ of a circle is called a "one-degree angle," and can be used to measure angles.

CCSS.MATH.CONTENT.4.MD.C.5.B

An angle that turns through n one-degree angles is said to have an angle measure of n degrees.

CCSS.MATH.CONTENT.4.MD.C.6

Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.

- How everyday materials are created, their origins and how they are obtained
- Changes of state based on temperature variation
- Differences between strong and weak forces and some consequences
- Sources of heat and light and how they are used by human beings
- Analyze characteristics of a diet and compares it to own eating habits
- Recognizes causes and effects of water, soil, and air pollution
- Solar system: eclipse and moon phases, relationship between moon, earth, and sun
- Characteristics of live beings and classifying
- Understand time can be measured by repetitive events and its relationship to live beings

- Mexican territory and political divisions
- Maps to obtain and learn information about territory
- Types of landscapes
- Characteristics of relief forms in Mexico & the spatial distributions dif types of maps
- Analyzes the distribution of water and its importance
- Relates the distribution of climates with the diversity of fauna and flora
- Natural disasters
- Characteristics and distribution of population, rural/urban
- Cultural diversity in Mexico
- Distribution and importance of natural resources
- Economy in mexico
- Transportation, communication, tourism in mexico
- Factors that influence the quality of life in mexico
- Assumes responsibility in the prevention and mitigation of environmental issues local and national
- Preservation of biodiversity of national territory